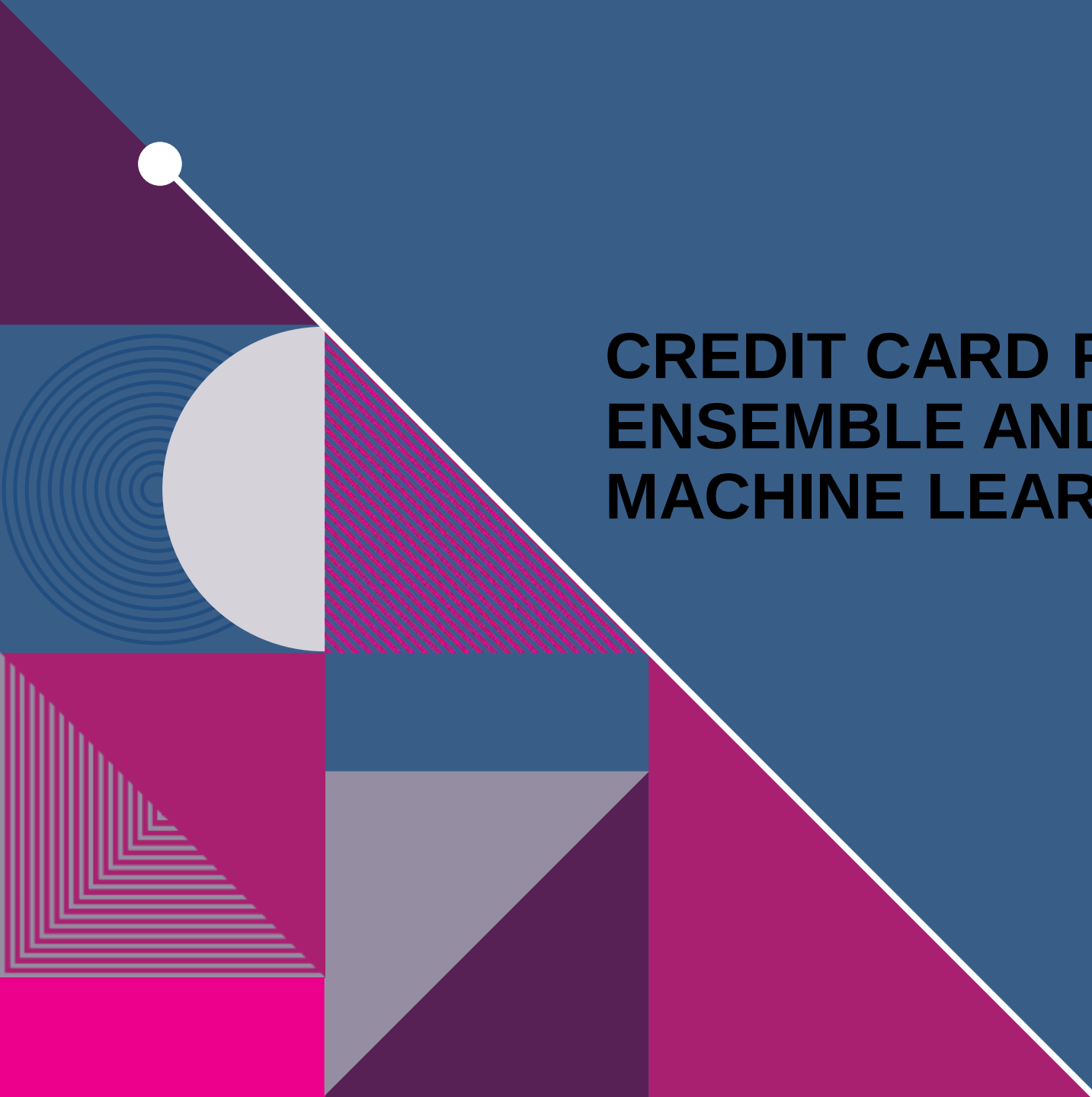
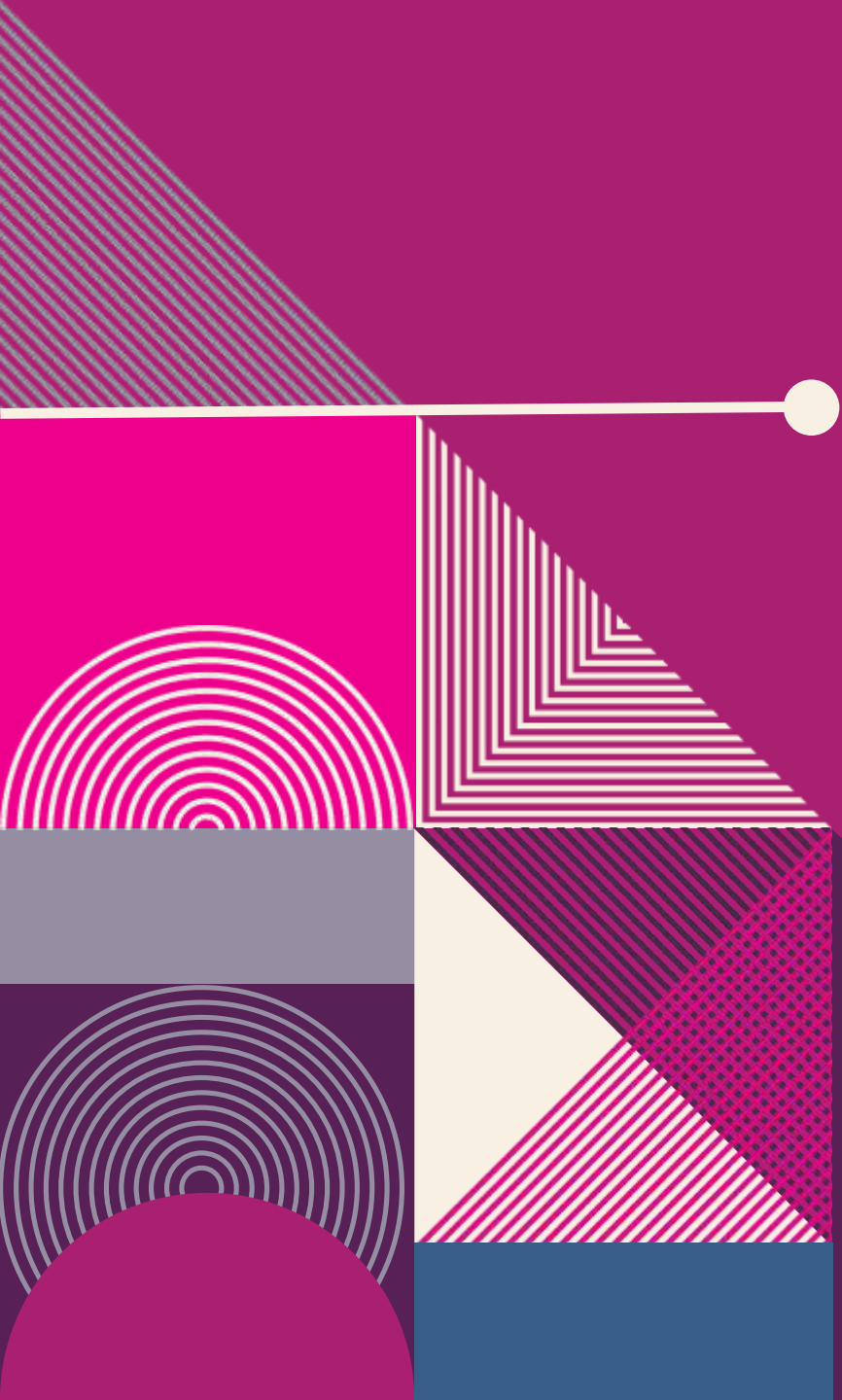




CREDIT CARD FRAUD DETECTION USING ENSEMBLE AND DISTANCE-BASED MACHINE LEARNING MODELS



PROJECT OVERVIEW

- Credit card fraud is a major financial security problem
- Fraudulent transactions are rare but costly
- Manual detection is difficult due to large data volume
- Machine Learning helps automate fraud detection
- System classifies transactions as fraudulent or legitimate
- Improves detection speed and accuracy

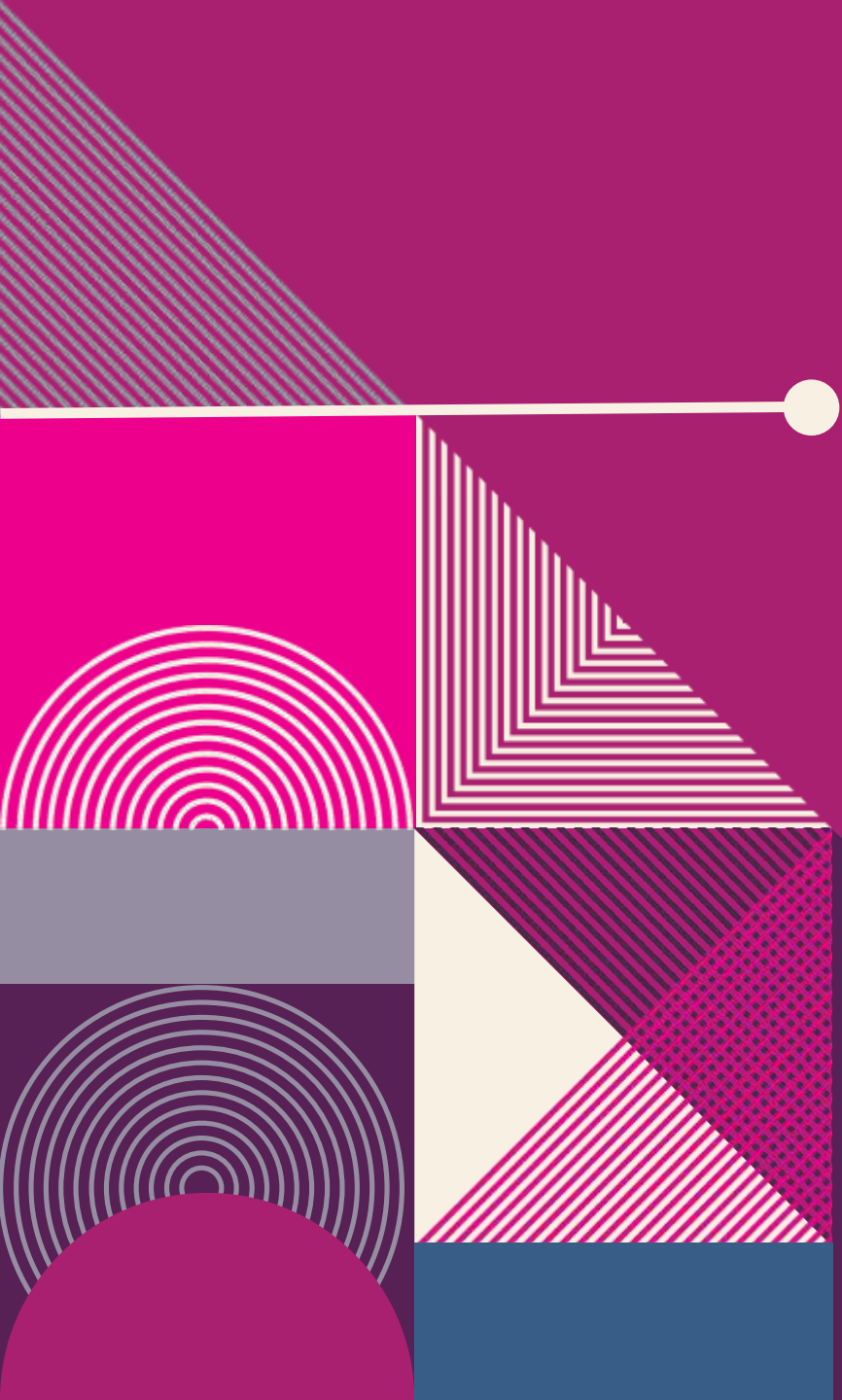
ALGORITHMS & TOOLS

Algorithms:

- Logistic Regression — Baseline binary classifier
- K-Nearest Neighbors (KNN) — Distance-based detection
- Random Forest — Ensemble of decision trees

Tools:

- Python
- Scikit-learn
- Pandas & NumPy
- Matplotlib / Seaborn



FEATURES AND CONCLUSION

Features:

- Automatic fraud detection system
- Handles large transaction datasets
- Combines multiple ML models for better performance
- Reduces false positives
- Identifies suspicious patterns quickly

Conclusion:

- Ensemble and distance-based models improve accuracy
- Enables real-time fraud prevention
- Helps financial institutions reduce losses
- Provides reliable and scalable solution

An abstract geometric design on the left side of the slide. It features a dark blue background with various geometric shapes and patterns. A white circle is positioned near the top left. Below it, a light blue semi-circle is visible. To the right of the semi-circle, there is a pattern of concentric circles. Further right, there is a pattern of parallel lines. The design is composed of several overlapping shapes in shades of blue, purple, and pink.

THANK YOU